Recitation 2

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Rutgers University

Question 1:

(a) True. Consider $B = 1 \subseteq R$. Then

$$\inf(B) = 1 = \sup(B).$$

Therefore,

$$\inf B \geq \sup B.$$

(b) False, every non-empty finite set will contain both its supremum and infimum. Thus a finite set cannot contain its infimum but fail to contain its supremum.

(c) Define

$$B = (0, 1] \cap \mathbb{Q}.$$

Then B is a bounded nonempty subset of $\mathbb{Q}$, where

$$\sup B = 1 \in B,$$

and

$$\inf B = 0 \notin B.$$

Question 2:

(a) The axiom of completeness says that every nonempty set in $\mathbb{R}$ that is bounded above has a supremum.

Similarly, for an ordered field F to be complete, every non-empty subset S of F that is bounded above must have a supremum.

(b) Let

$$S = \mathbb{Q} \cap (0, \pi).$$

Suppose, for sake of contradiction, that S has a supremum in the ambient ordered field of $\mathbb{Q}$, say

$$s = \sup S.$$

Notice that $\pi > x$ for every $x \in S$, so

$$s \leq \pi$$

Since $s \in \mathbb{Q}$ and $\pi \notin \mathbb{Q}$,

$$s < \pi$$

Furthermore, $\mathbb{Q}$ is dense in $\mathbb{R}$, so there must exist some $q \in \mathbb{Q}$ such that

$$s < q < \pi.$$

Since $s \in S$, it follows that $0 < s < \pi$. So

$$0 < q < \pi,$$

and $q \in S$. But $q > s$, which contradicts our assumption that s is an upper bound on S. Therefore, S has no supremum in $\mathbb{Q}$.

If we instead choose the ambient ordered field to be $\mathbb{R}$, then π is still an upper bound on S since every $x \in S$ satisfies

$$\pi \geq x.$$

For sake of contradiction, suppose there exists a $u < \pi$ such that

$$u \geq x$$

for every $x \in S$. Then, by density of $\mathbb{Q}$, there exists a $q \in \mathbb{Q}$ such that

$$u < q < \pi$$

But, then $q \in S$, so u cannot be an upper bound on S. Therefore π must be the least upper bound.

This proves that $\mathbb{Q}$ is not complete, because even though $S \subseteq \mathbb{Q}$ is nonempty and bounded above, it has no supremum in $\mathbb{Q}$

(c) Let

$$Q_1 = \mathbb{Q} \cap (0, \pi),\ Q_2 = \mathbb{Q} \cap (\pi, 100)$$

For every $a \in Q_1$

$$a < \pi$$

and for every $b \in Q_2$

$$b > \pi.$$

Thus,

$$a < \pi < b,$$

so

$$a < b.$$

Suppose, for contradiction, that there exists some $c \in \mathbb{Q}$ such that

$$a \leq c \leq b$$

for every $a \in Q_1$ and every $b \in Q_2$. Since $c \in \mathbb{Q}$ and $\pi \notin \mathbb{Q}$, either $c < \pi$ or $c > \pi$.

(Case 1) If $c < \pi$, then density of $\mathbb{Q}$ says there exists some $q \in \mathbb{Q}$ such that

$$c < q < \pi.$$

So $q \in Q_1$. But $q > c$, contradicting $c \geq a$ for every $a \in Q_1$.

(Case 2) If $c > \pi$, then density of $\mathbb{Q}$ says there exists some $q \in \mathbb{Q}$ such that

$$\pi < q < c.$$

So $q \in Q_2$. But $q < c$, contradicting $c \leq b$ for every $b \in Q_2$.

Since both cases are impossible, this c cannot exist in $\mathbb{Q}$. Therefore, there is no single $c \in \mathbb{Q}$ such that

$$a \leq c \leq b$$

for every $a \in Q_1$ and every $b \in Q_2$.

Question 3:

(a) Let $X, Y \subseteq \mathbb{R}$ be nonempty. Assume that X is bounded above, Y is bounded below,

$$x \leq y, \text{ for every } x \in X \text{ and for every } y \in Y.$$

That is, every y is an upper bound on X. Since $\sup X$ is the least upper bound, it follows that

$$\sup X \leq y, \text{ for every } y \in Y.$$

That is, $\sup X$ is a lower bound on Y. Since $\inf Y$ is the greatest lower bound, it follows that

$$\sup X \leq \inf Y.$$

(b) Continuing from part (a), assume

$$X \cup Y = \mathbb{R}.$$

From part (a), we have

$$\sup X \leq \inf Y.$$

For sake of contradiction, suppose that the strict inequality holds. So

$$\sup X < \inf Y.$$

Then, the midpoint would be

$$m = \frac{\sup X + \inf Y}{2}.$$

So

$$\sup X < m < \inf Y$$

Since $X \cup Y = \mathbb{R}$, either $m \in X$ or $m \in Y$.

(Case 1) If $m \in X$, then by definition of supremum,

$$\sup X \geq m.$$

However, this contradicts that $\sup X < m$.

(Case 2) If $m \in Y$, then by definition of infimum,

$$\inf Y \leq m$$

However, this contradicts that $\inf Y > m$.

Both cases lead to contradictions. Since $m \in \mathbb{R}$ and $X \cup Y = \mathbb{R}$, this is impossible. The strict inequality does not hold, so from $\sup X \leq \inf Y$ we get

$$\sup X = \inf Y.$$

Question 4

Let $A \subseteq \mathbb{R}$ satisfy the relative Nested Interval Property, and $E \subseteq A$ be nonempty and bounded above by some element of A. Define

$$U = \{u \in A : u \text{ is an upper bound for } E\}$$

and

$$L = A \setminus U.$$

(a) Assume, for the sake of contradiction, that U has no least element. By definition, E is bounded above by some element of A. So there exists some $u \in A$ that is an upper bound on E. Then $u \in U$, and so

$$U \neq \emptyset.$$

Take some $e \in E$. For sake of contradiction, suppose $e \in U$. Then by definition of U,

$$e \leq u \text{ for every } u \in U.$$

That is, e is the least element of U. However, this contradicts our assumption that U has no least element. So $e \notin U$, which means $e \in L$. Thus,

$$L \neq \emptyset.$$

Take some $\ell \in L$ and $u \in U$. Since $\ell \notin U$, ℓ is not an upper bound on E, there exists some $e \in E$ such that

$$e > \ell.$$

Since u is an upper bound on E,

$$u \geq e.$$

Together,

$$\ell < e \leq u.$$

Thus,

$$\ell < u \text{ for every } \ell \in L \text{ and } u \in U.$$

Suppose, for sake of contradiction, that there exists some maximum ℓ_0 in L.

Since $\ell_0 \in L$, it is not an upper bound for E. So there exists some $e \in E$ such that

$$e > \ell_0.$$

Since ℓ_0 is the largest element of L and e is greater, $e \notin L$. Furthermore, because $e \in E \subseteq A$, it follows that $e \in U$. With $e \in U$ and $e \in E$, it follows that

$$e \leq u \text{ for every } u \in U.$$

That is, e is the least element of U. However, this contradicts our assumption that U has no least element. Therefore, it must be that L has no largest element.

(b) Since L is nonempty and bounded above, and U is nonempty and bounded below, completeness of $\mathbb{R}$ gives

$$\alpha = \sup L \qquad \text{and} \qquad \beta = \inf U.$$

Because $\alpha = \sup L$ and L has no largest element, we may recursively choose $(a_n) \subseteq L$ so that

$$a_1 \leq a_2 \leq \cdots$$

and

$$\alpha - \frac{1}{n} < a_n \leq \alpha.$$

Similarly, because $\beta = \inf U$ and U has no least element, we may choose $(b_n) \subseteq U$ so that

$$b_1 \geq b_2 \geq \cdots$$

and

$$\beta \leq b_n < \beta + \frac{1}{n}.$$

(c) Define

$$I_n = [a_n, b_n] \cap A.$$

Recall from part (a) that

$$\ell < u \text{ for every } \ell \in L \text{ and every } u \in U.$$

Since $(a_n) \subseteq L$ and $(b_n) \subseteq U$, it follows that

$$a_n < b_n \text{ for every } n \in \mathbb{N}.$$

Furthermore, because (a_n) is increasing while (b_n) is decreasing, each interval will contain the next. That is,

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots.$$

Thus, by the relative Nested Interval Property,

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

Choose some

$$x \in \bigcap_{n=1}^{\infty} I_n.$$

Then for every $n \in \mathbb{N}$,

$$a_n \leq x \leq b_n.$$

Suppose, for sake of contradiction, that $x < \alpha$. Then

$$\alpha - x > 0.$$

By the Archimedean Property, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < \alpha - x.$$

Thus,

$$x < \alpha - \frac{1}{n} < a_n.$$

But, this contradicts $a_n \leq x$. Therefore,

$$x \geq \alpha.$$

Similarly, suppose, for sake of contradiction, $x > \beta$ Then

$$x - \beta > 0.$$

By the Archimedean Property, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < x - \beta.$$

Thus,

$$b_n < \frac{1}{n} + \beta < x.$$

But, this contradicts $x \leq b_n$. Therefore,

$$x \leq \beta$$

Together

$$\alpha \leq x \leq \beta.$$

Hence,

$$x \in A \cap [\alpha, \beta].$$

Since $A = L \cup U$, either $x \in L$ or $x \in U$.

(Case 1) Say $x \in L$. Then since α is the supremum of L,

$$x \leq \alpha.$$

But since $x \in [\alpha, \beta]$,

$$x \geq \alpha.$$

So $x = \alpha$. Since $x \in L$, this means $\alpha \in L$. But this contradicts the fact that L has no largest element.

(Case 2) Say $x \in U$. Then since $\beta = \inf U$,

$$x \geq \beta.$$

Since $x \in [\alpha, \beta]$,

$$x \leq \beta.$$

So $x = \beta$. Since $x \in U$, this means $\beta \in U$, contradicting the fact that U has no least element.

Thus, neither case is possible. But $x \in A$, so one of them must hold. This is a contradiction. Hence, our original assumption that U has no least element must be

false. Therefore, U has a least element, say, s. Since U is the set of upper bounds of E, s must be the least upper bound of E in A. That is,

$$s = \sup_A E.$$

Since $E \subseteq A$ was arbitrary, every nonempty subset of A that is bounded above by an element of A has a least upper bound belonging to A.

Therefore, by definition of completeness, we conclude that A is complete.

Question 5

(a) Define $f : \mathbb{N} \cup \{0\} \to \mathbb{N}$ as $f(x) = x + 1$

Injective: Suppose $f(y) = f(x)$. Then

$$x + 1 = y + 1.$$

So

$$x = y.$$

Thus, $f(x)$ is injective.

Surjective: Let $y \in \mathbb{N}$. Choose $x = y - 1$. Then

$$f(x) = y - 1 + 1 = y.$$

Thus, $f(x)$ is also surjective.

Therefore $f(x)$ is bijective.

(b) Define

$$f : \{0\} \cup D \to D$$

as

$$f(x) = \begin{cases} \frac{1}{2}, \text{ if } x = 0 \\ \frac{1}{2^{n+1}}, \text{ if } x = \frac{1}{2^n} \end{cases}$$

Injectivity:

(Case 1) Suppose $f(x) = f(y)$.

If $f(x) = \frac{1}{2}$, then the only possible input is $x = 0$, and similarly $y = 0$.

(Case 2) If $x = \frac{1}{2^n}$. Suppose $f(x) = f(y)$. Then by definition of f,

$$x = y = \frac{1}{2^n}.$$

Thus, f is injective in all cases.

Surjectivity:

Let $y \in D$. Then $y = \frac{1}{2^n}$ for some $n \in \mathbb{N}$. If $n = 1$, then $y = \frac{1}{2}$. So we can just choose $x = 0$. Then

$$f(0) = \frac{1}{2} = y.$$

If $n \geq 2$, then $y = \frac{1}{2^n}$. So we choose

$$x = \frac{1}{2^{n-1}}.$$

Then

$$f(x) = \frac{1}{2^n} = y.$$

Thus, f is surjective. Therefore, f is a bijective function.

(c) Define

$$f : [0, 1) \to (0, 1)$$

as

$$f(x) = \begin{cases} \frac{1}{2}, & x = 0, \\ \frac{1}{2^{n+1}}, & x = \frac{1}{2^n} \text{ for some } n \in \mathbb{N}, \\ x, & \text{otherwise.} \end{cases}$$

Injectivity:

(Case 1) Suppose $f(x) = f(y)$.

If $f(x) = \frac{1}{2}$, then the only possible input is $x = 0$, and similarly $y = 0$.

(Case 2) If $x = \frac{1}{2^n}$. Suppose $f(x) = f(y)$. Then by definition of f,

$$x = y = \frac{1}{2^n}.$$

(Case 3) For all the other x values, suppose $f(x) = f(y)$. Then $x = y$.

Thus, f is injective in all cases.

Surjectivity:

Let $y \in (0, 1)$.

(Case 1) If $y = \frac{1}{2}$. Choose $x = 0$. Then

$$f(0) = y$$

(Case 2) If $y = \frac{1}{2^n}$. Choose $x = \frac{1}{2^{n-1}}$. Then

$$f(x) = \frac{1}{2^{n-1+1}} = y.$$

(Case 3) If y is any other value in $(0, 1)$, just choose $x = y$. Then

$$f(x) = y.$$

Thus, f is surjective. Therefore, f is a bijective function.

(d) Define

$$g : [0, 1] \to [0, 1)$$

as

$$g(x) = \begin{cases} \frac{1}{2}, & x = 1, \\ 1 - \frac{1}{2^{n+1}}, & x = 1 - \frac{1}{2^n} \text{ for some } n \in \mathbb{N}, \\ x, & \text{otherwise.} \end{cases}$$

Injectivity: (Case 1) Suppose $f(x) = f(y)$.

If $f(x) = \frac{1}{2}$, then the only possible input is $x = 1$, and similarly $y = 1$.

(Case 2) If $y = 1 - \frac{1}{2^n}$. Suppose $g(x) = g(y)$. Then, by definition of g,

$$x = y = 1 - \frac{1}{2^n}$$

(Case 3) For all the other x values, suppose $g(x) = g(y)$. Then $x = y$.

Thus, g is injective in all cases.

Surjectivity:

Let $y \in [0, 1)$.

(Case 1) If $y = \frac{1}{2}$. Choose $x = 1$. Then

$$g(1) = \frac{1}{2} = y$$

(Case 2) If $y = 1 - \frac{1}{2^n}$. Choose $x = 1 - \frac{1}{2^{n-1}}$. Then

$$g(x) = 1 - \frac{1}{2^{n-1+1}} = y.$$

(Case 3) If y is any other value in $[0, 1)$, just choose $x = y$. Then

$$g(x) = y.$$

Thus, g is surjective.

That is, g is bijective.

Recall from part (c), we concluded that the function

$$f : [0, 1) \to (0, 1)$$

is bijective as well.

Hence, the composition of the two functions

$$f \circ g : [0, 1] \to (0, 1)$$

must also be a bijection. Therefore,

$$[0, 1] \sim (0, 1)$$